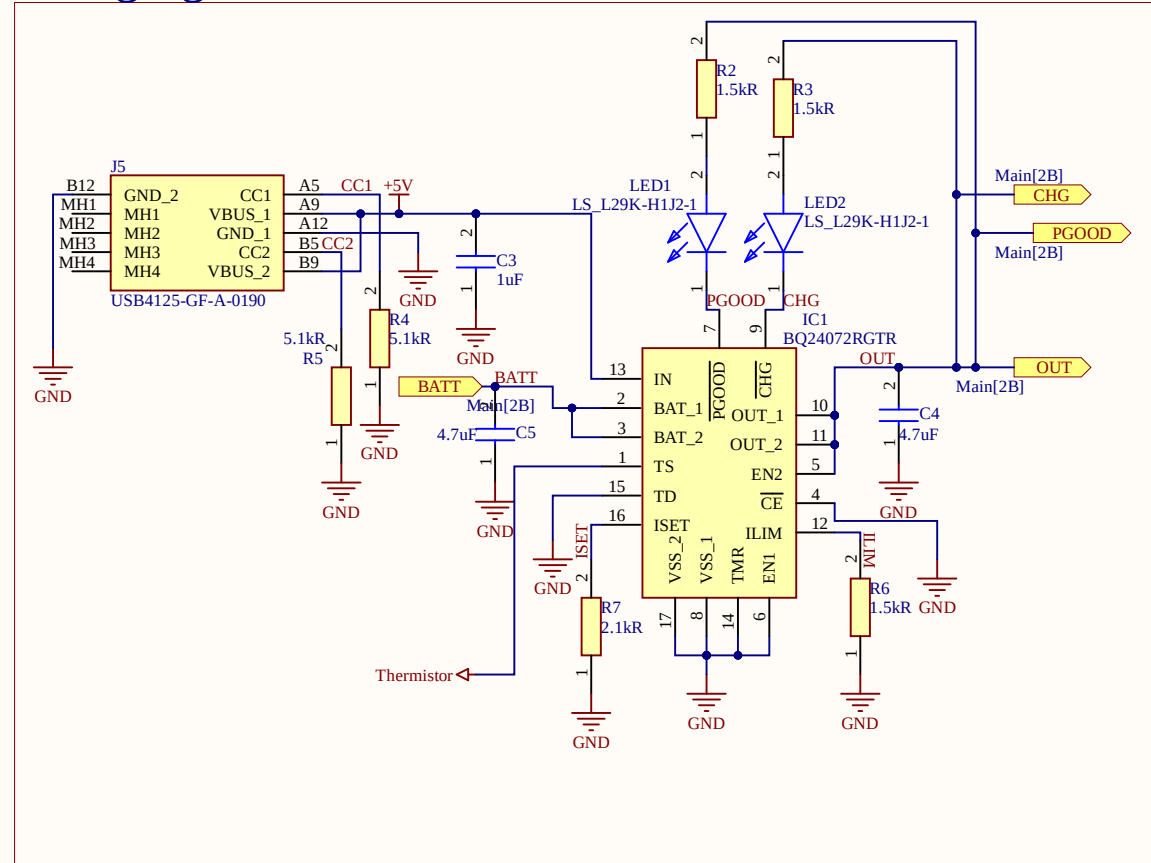


Charging



Max current during charging is 425mA
 Max current during gamepad working is 1A
 Max allowed temperature of battery pack is 45 C
 PGOOD LED inform about type C voltage
 CHG LED inform about battery charging, it will go out after battery reach 100%

10.2.2.2 Calculations

10.2.2.2.1 Program the Fast Charge Current (ISET):

$$R_{ISET} = K_{ISET} / I_{CHG}$$

$K_{ISET} = 890 \text{ A}\Omega$ from the electrical characteristics table.

$$R_{ISET} = 890 \text{ A}\Omega / 0.8 \text{ A} = 1.1125 \text{ k}\Omega$$

Select the closest standard value, which for this case is 1.13 k Ω . Connect this resistor between ISET (pin 16) and V_{SS}.

10.2.2.2.2 Program the Input Current Limit (ILIM)

$$R_{ILIM} = K_{ILIM} / I_{L_MAX}$$

$K_{ILIM} = 1550 \text{ A}\Omega$ from the electrical characteristics table.

$$R_{ISET} = 1550 \text{ A}\Omega / 1.3 \text{ A} = 1.192 \text{ k}\Omega$$

Select the closest standard value, which for this case is 1.18 k Ω . Connect this resistor between ILIM (pin 12) and V_{SS}.

10.2.2.2.3 Program the Termination Current Threshold (I_{TERM}) (BQ24074 only)

$$R_{ITERM} = I_{TERM} \times R_{ISET} / 0.030$$

$$R_{ISET} = 1.13 \text{ k}\Omega \text{ from the above calculation.}$$

$$R_{ITERM} = 110 \text{ mA} \times 1.13 \text{ k}\Omega / 0.030 = 4.143 \text{ k}\Omega$$

Select the closest standard value, which for this case is 4.12 k Ω . Connect this resistor between ITERM (pin 15) and V_{SS}. Note that when in USB100 mode (EN1 = EN2 = V_{SS}), the termination threshold is 1/3 of the normal threshold.

10.2.2.2.4 Program 6.25-hour Fast-Charge Safety Timer (TMR)

$$R_{TMR} = I_{MAXCHG} / (10 \times K_{TMR})$$

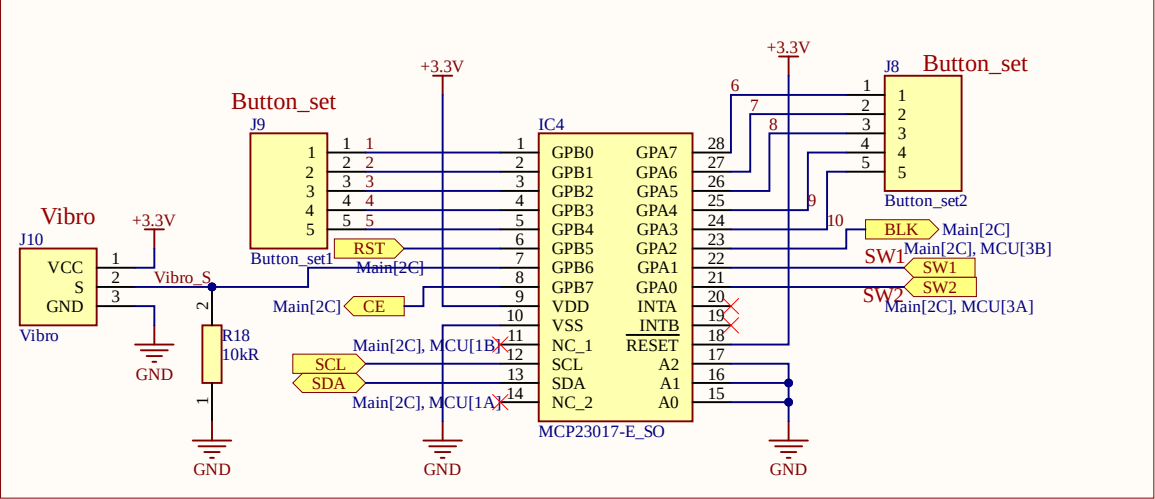
$K_{TMR} = 48 \text{ s/k}\Omega$ from the electrical characteristics table.

$$R_{TMR} = (6.25 \text{ hr} \times 3600 \text{ s/hr}) / (10 \times 48 \text{ s/k}\Omega) = 46.8 \text{ k}\Omega$$

Select the closest standard value, which for this case is 46.4 k Ω . Connect this resistor between TMR (pin 14) and V_{SS}.

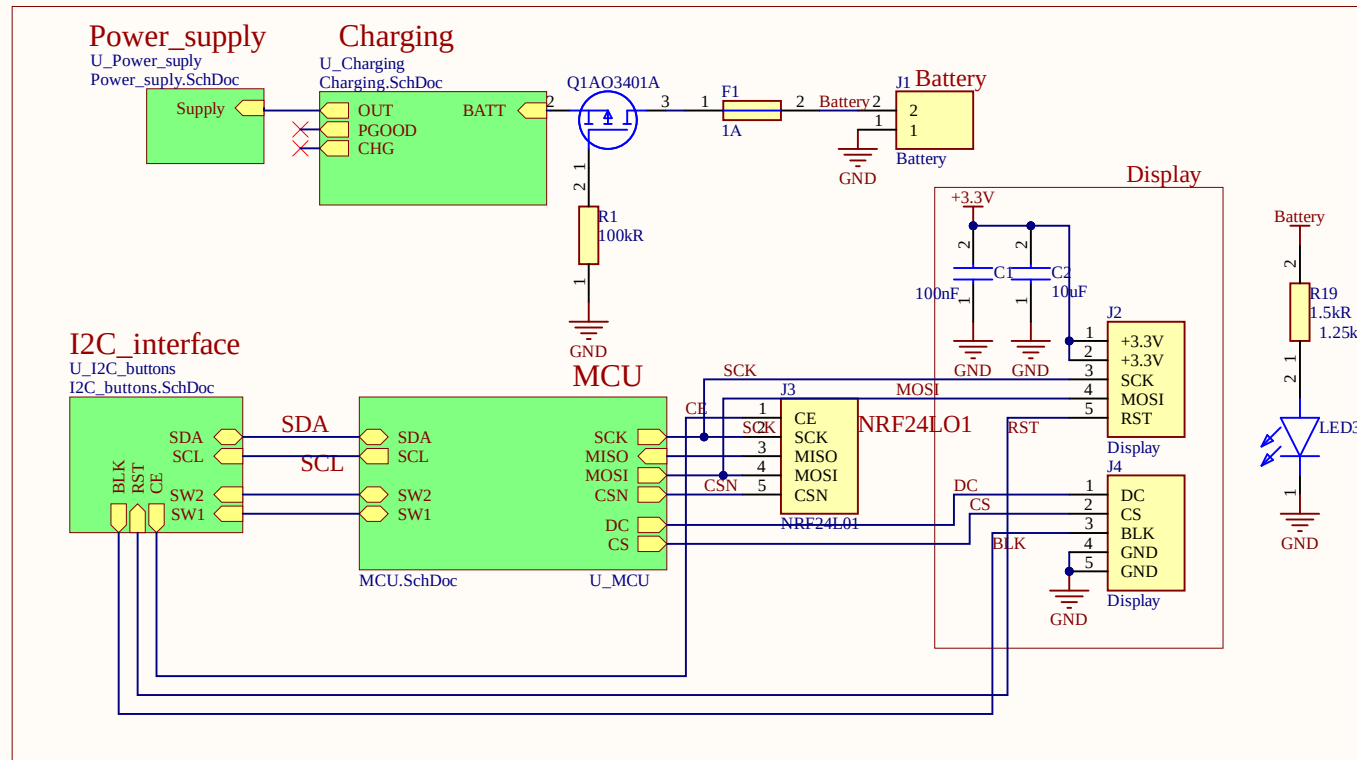
Title Main sheet		
Size A4	Number	Revision
Date: 9.16.2026	Sheet of	
File: C:\Users\...\Charging.SchDoc	Drawn By: Stanislav Kravtsov	

I2C_buttons_bus



Title I2C buss		
Size A4	Number	Revision
Date: 9.16.2026	Sheet of	
File: C:\Users\...\I2C_buttons.SchDoc	Drawn By: Stanislav Kravtsov	

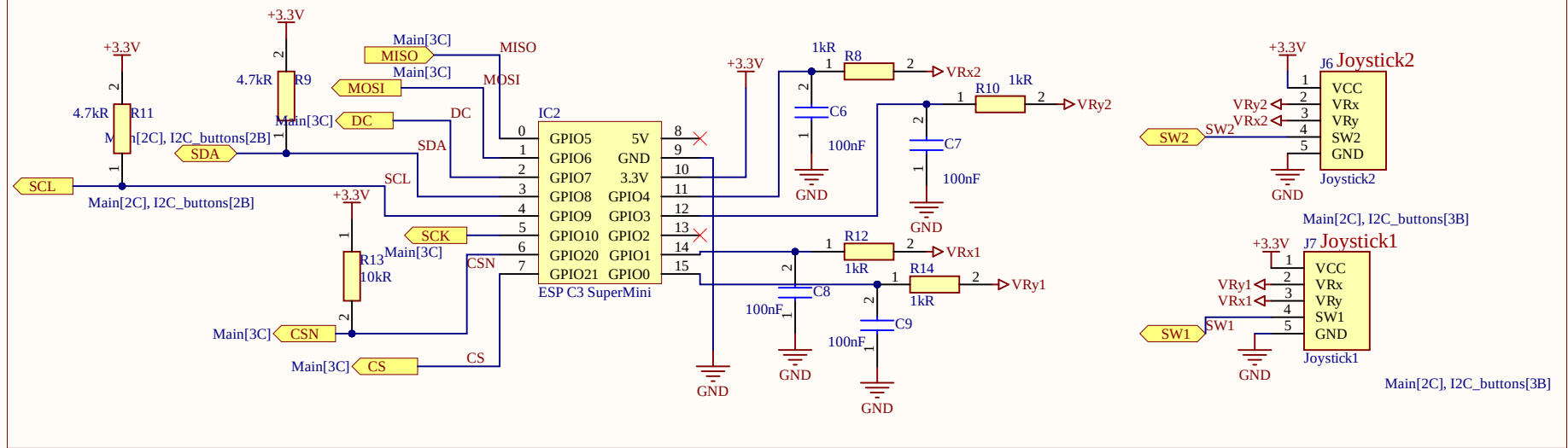
Gamepad



$$R19 = U_r/I = 2.5/2 \cdot 10^{-3} = 1250 \text{ Ohm}$$

Title Main sheet		
Size A4	Number	Revision
Date: 9.16.2026	Sheet of	
File: C:\Users\...\Main.SchDoc	Drawn By: Stanislav Kravtsov	

MCU ESP C3 Super mini

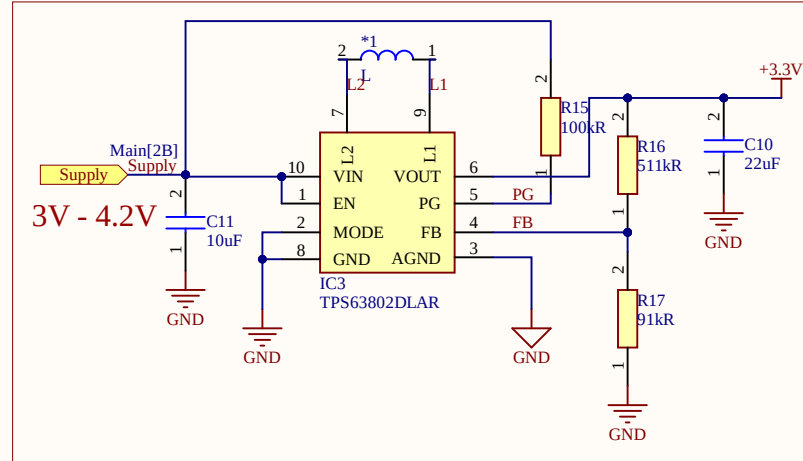


PIN	Function	Device
GPIO0	VRy1	Joystick1Y
GPIO1	VRx1	Joystick1X
GPIO2	-(Programing)	-
GPIO3	VRy2	Joystick2Y
GPIO4	VRx2	Joystick2X
GPIO5	MISO	NRF24LO1
GPIO6	MOSI	Display/NRF24LO1
GPIO7	DC	Display
GPIO8	SDA	I2C_bus
GPIO9	SCL	I2C_bus
GPIO10	SCK	Display/NRF24LO1
GPIO20	CSN	NRF24LO1
GPIO21	CS	Display
5V	-	-
3.3V	3.3V	-
GND	GND	-

Title MCU		
Size A4	Number	Revision
Date:	9.16.2026	Sheet of
File:	C:\Users\...\MCU.SchDoc	Drawn By: Stanislav Kravtsov

BUCK-BOOST

2A max out



$$U_{out} = U_{fb} (1 + R1/R2) = 0.5 (1 + 511/91) = 3.3077V$$

$$\text{Working time} = 850\text{mAh} / 230\text{mA} = 3.69\text{h}$$

Device	Current
Joysticks	5mA
ESP c3 mini	100mA
Vibromotor	50mA
TFT display	60mA
NRF24L01	15mA
Total	230mA

Title DC-DC Buck-boost		
Size A4	Number	Revision
Date:	9.16.2026	Sheet of
File:	C:\Users\...\Power_suply.SchDoc	Drawn By: Stanislav Kravtsiv